

EVENES HARSTAD NARVIK, Norway

WMO: 011830

Lat: 68.491N

Lon: 16.678E

Elev: 26

StdP: 101.02

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-15.8	-20.9	0.6	-16.0	-18.9	0.7	-14.5	14.1	-2.3	12.2	0.4	1.1	10	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	23.1	15.6	20.9	14.6	18.9	13.5	16.8	21.6	15.3	19.5	14.2	17.8	3.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	10.3	19.0	13.3	9.5	16.8	12.3	9.0	15.6	47.1	22.0	42.8	19.7	39.9	17.8	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
10.4	8.9	7.7	DB	-21.2	26.8	1.5	3.1	-22.2	29.0	-23.1	30.8	-23.9	32.5	-25.0	34.8	(4)
			WB	-21.2	18.4	1.6	1.8	-22.4	19.7	-23.3	20.7	-24.2	21.7	-25.3	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-4.8	-4.3	-2.4	1.8	6.9	10.2	13.5	12.4	8.8	3.9	-0.3	-2.5
	DBStd	7.49	5.25	5.12	4.38	3.10	3.66	3.25	3.21	2.70	3.08	3.81	4.59	4.89
	HDD10.0	2591	457	401	386	245	112	35	3	8	57	193	308	388
	HDD18.3	5367	716	634	644	495	356	243	156	186	285	449	558	646
	CDD10.0	272	0	0	0	0	14	42	111	81	22	2	0	0
	CDD18.3	7	0	0	0	0	0	1	6	1	0	0	0	0
	CDH23.3	77	0	0	0	0	6	9	57	5	0	0	0	0
	CDH26.7	13	0	0	0	0	1	1	11	1	0	0	0	0
Wind	WSAvg	3.3	3.5	3.6	3.4	3.3	3.7	3.5	3.3	3.0	3.0	3.0	3.0	3.3
Precipitation	PrecAvg	1281	120	113	116	68	72	69	90	100	147	140	111	133
	PrecMax	1578	313	225	268	159	114	127	182	231	308	266	225	313
	PrecMin	1058	0	20	17	9	15	17	21	44	38	30	8	0
	PrecStd	161	77	51	66	37	30	28	50	43	68	57	65	76
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	6.0	7.6	14.2	22.8	24.2	28.0	23.4	19.4	14.9	9.2	7.5
		MCWB	5.4	4.1	3.7	7.8	13.3	14.8	18.8	16.3	13.0	9.9	7.2	5.9
	2%	DB	5.1	5.0	5.7	10.8	18.8	21.0	24.6	20.9	17.0	12.1	8.1	6.2
		MCWB	3.7	3.3	3.2	5.8	11.1	13.6	16.9	15.1	12.3	8.5	6.3	4.4
	5%	DB	4.0	4.0	4.6	8.0	15.2	18.2	22.0	18.9	15.0	10.2	7.0	4.9
		MCWB	2.5	2.6	2.4	4.6	9.3	12.0	15.6	14.1	11.2	8.2	5.1	3.0
	10%	DB	2.2	2.9	3.5	6.5	12.8	15.9	19.8	17.1	13.2	9.1	5.8	3.5
		MCWB	0.8	1.5	1.7	3.7	7.9	11.1	14.7	13.5	10.6	7.4	3.9	2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	5.1	4.6	8.2	13.6	15.5	19.6	17.8	13.9	10.5	8.1	6.7
		MCDB	6.1	5.5	6.1	13.4	22.4	22.4	26.4	22.2	17.3	12.7	8.9	7.0
	2%	WB	3.9	3.5	3.7	6.4	11.2	14.1	17.8	15.7	12.7	9.6	6.9	4.6
		MCDB	4.8	4.3	5.0	8.8	17.7	19.5	23.7	19.4	15.5	11.2	7.7	5.8
	5%	WB	2.5	2.7	2.7	5.1	9.7	12.8	16.2	14.8	11.9	8.6	5.3	3.2
		MCDB	3.6	3.8	4.1	7.4	14.9	17.0	20.9	18.2	14.3	10.2	6.4	4.5
	10%	WB	1.1	1.5	1.8	4.1	8.3	11.6	14.9	13.7	11.1	7.5	4.0	2.0
		MCDB	2.0	2.6	3.2	6.2	11.8	15.1	18.7	16.5	13.1	9.0	5.3	3.3
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.5	7.1	6.7	7.1	6.4	6.9	6.9	6.5	5.2	5.2	5.8
		MCDBR	5.5	5.4	5.4	8.6	11.8	11.3	12.3	10.7	9.0	5.5	4.9	5.5
		MCWBR	4.7	4.7	3.9	5.3	5.7	5.4	5.9	5.7	5.2	4.0	3.9	4.7
	5% WB	MCDBR	5.3	5.2	5.2	7.4	11.0	9.5	11.4	9.5	7.5	5.0	4.8	5.3
		MCWBR	4.9	4.9	4.1	5.1	5.5	5.0	5.9	5.4	4.7	4.1	4.4	4.8
Clear-Sky Solar Irradiance	taub	N/A	0.262	0.303	0.324	0.336	0.338	0.343	0.332	0.313	0.276	0.353	N/A	
	taud	N/A	2.106	2.181	2.285	2.428	2.495	2.510	2.528	2.551	2.278	2.655	N/A	
	Ebn at Noon	N/A	511	716	820	856	865	846	816	739	502	101	0	
	Edh at Noon	N/A	45	78	95	93	90	86	76	57	38	10	0	
All-Sky Solar Radiation	RadAvg	0.02	0.32	1.42	3.24	4.81	4.87	4.41	3.16	1.69	0.58	0.07	0.00	
	RadStd	0.00	0.07	0.23	0.27	0.40	0.66	0.48	0.30	0.18	0.10	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page